SanthoshRamesh123 / Developing-a-Neural-Network-Regression-Model

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96 lines (63 loc) · 2.42 KB

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Developing a Neural Network Regression Model

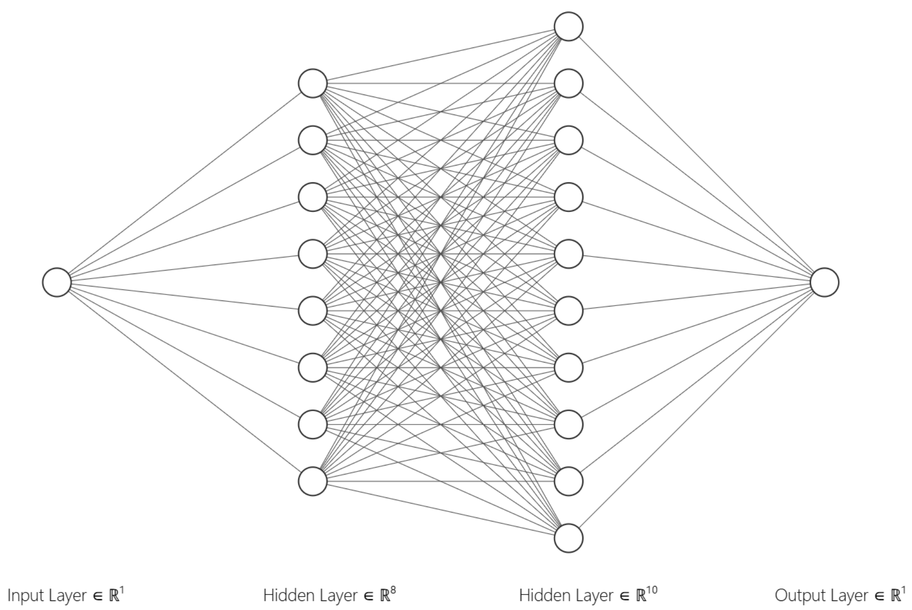
AIM

To develop a neural network regression model for the given dataset.

THEORY

Explain the problem statement

Neural Network Model



DESIGN STEPS

STEP 1:

Create your dataset in a Google sheet with one numeric input and one numeric output.

STEP 2:

Split the dataset into training and testing

STEP 3:

Create MinMaxScalar objects ,fit the model and transform the data.

STEP 4:

Build the Neural Network Model and compile the model.

STEP 5:

Train the model with the training data.

STEP 6:

Plot the performance plot

STEP 7:

Evaluate the model with the testing data.

STEP 8:

Use the trained model to predict for a new input value .

PROGRAM

```
class NeuralNet(nn.Module):
    def __init__(self):
        super().__init__()

        self.fc1 = nn.Linear(1, 8)
        self.fc2 = nn.Linear(8, 10)
        self.fc3 = nn.Linear(10, 1)

        self.relu = nn.ReLU()
        self.history = {'loss': []}

    def forward(self, x):
        x = self.relu(self.fc1(x))
        x = self.relu(self.fc2(x))
        x = self.fc3(x)
```

```
        return x


def train_model(ai_brain, X_train, y_train, criterion, optimizer, epochs=2000):
    for epoch in range(epochs):
        optimizer.zero_grad()
        loss=criterion(ai_brain(X_train),y_train)
        loss.backward()
        optimizer.step()


        ai_brain.history['loss'].append(loss.item())
        if epoch % 200 == 0:
            print(f'Epoch [{epoch}/{epochs}], Loss: {loss.item():.6f}')
```

Name:

SANTHOSH R

Register Number:

212224230249

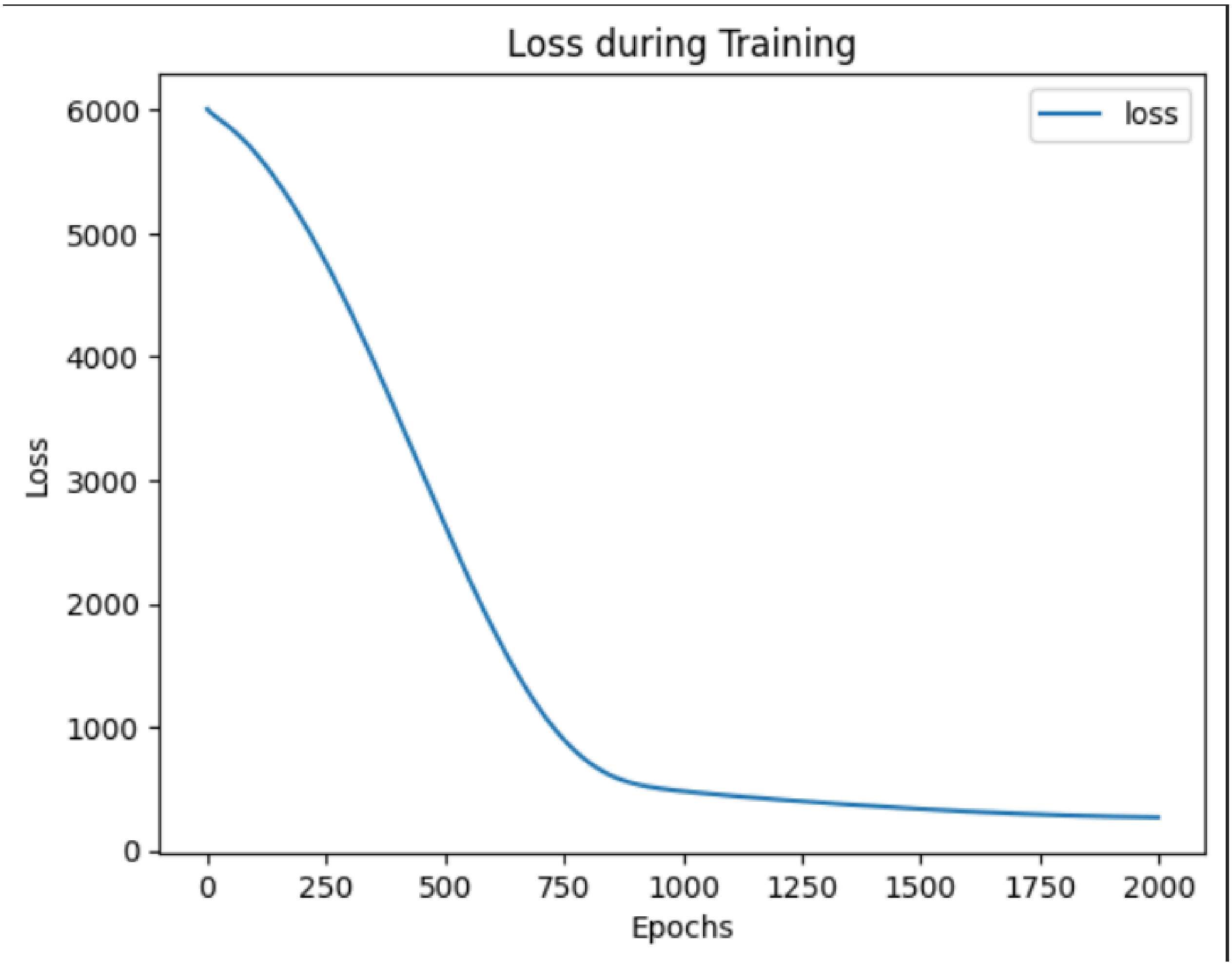
Dataset Information

INPUT	OUTPUT
1	90
2	80
3	85
4	95
5	91
6	50
7	65
8	75
9	87
10	69

OUTPUT

```
Epoch [0/2000], Loss: 6009.021973
Epoch [200/2000], Loss: 5106.371582
Epoch [400/2000], Loss: 3525.109863
Epoch [600/2000], Loss: 1801.952759
Epoch [800/2000], Loss: 718.833374
Epoch [1000/2000], Loss: 478.152100
Epoch [1200/2000], Loss: 411.184509
Epoch [1400/2000], Loss: 357.260437
Epoch [1600/2000], Loss: 313.923523
Epoch [1800/2000], Loss: 283.523804
```

Training Loss Vs Iteration Plot



New Sample Data Prediction

```
Prediction: 74.49366760253906
```

RESULT

Thus, a neural network regression model was successfully developed and trained using PyTorch.